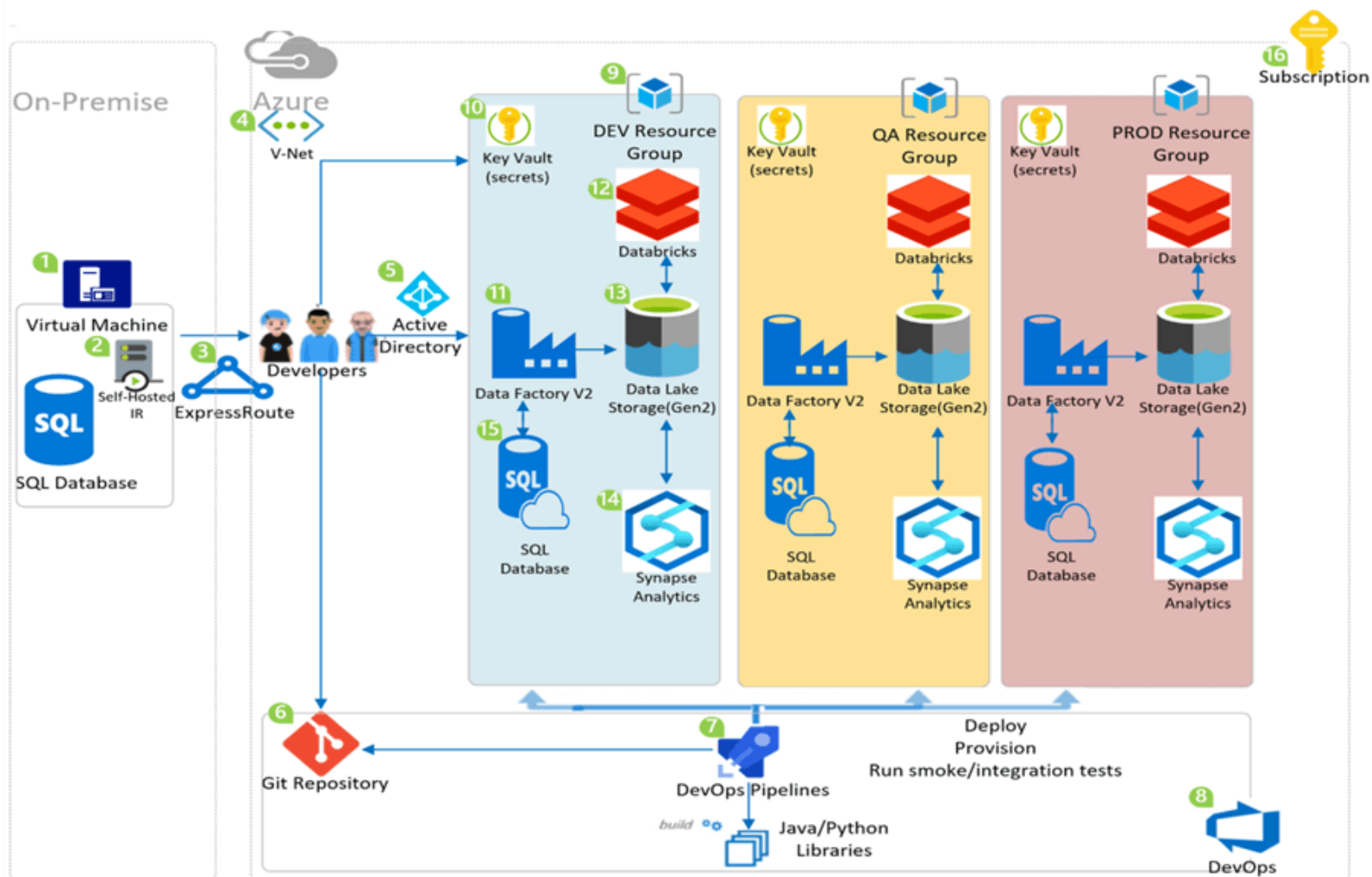




Nitya CloudTech
Dream.Achieve.Succeed

Data Engineering Interview Questions and Answers





Nitya CloudTech
Dream.Achieve.Succeed

Interviewer:

Your company uses Azure services to integrate data from multiple sources and create analytical dashboards.

Suppose you need to ingest and process 2 TB of data daily from three different sources: SQL Server, an SFTP server, and REST APIs. How would you design the data pipeline?





Nitya CloudTech

Dream.Achieve.Succeed

I would use Azure Data Factory (ADF) as the primary tool to orchestrate the pipeline:

- Use Copy Activity in ADF to ingest data from SQL Server, SFTP, and REST APIs.
- Set up a self-hosted integration runtime for on-premises SQL Server connectivity.
- Land the ingested data in Azure Data Lake Storage Gen2 for staging.
- Use Mapping Data Flows or Azure Databricks for data transformation, including cleansing, deduplication, and enrichment.
- Load the transformed data into Azure Synapse Analytics for analytical querying and reporting

Interviewer:

How would you optimize this pipeline to handle potential bottlenecks, such as high latency or failures during data ingestion?



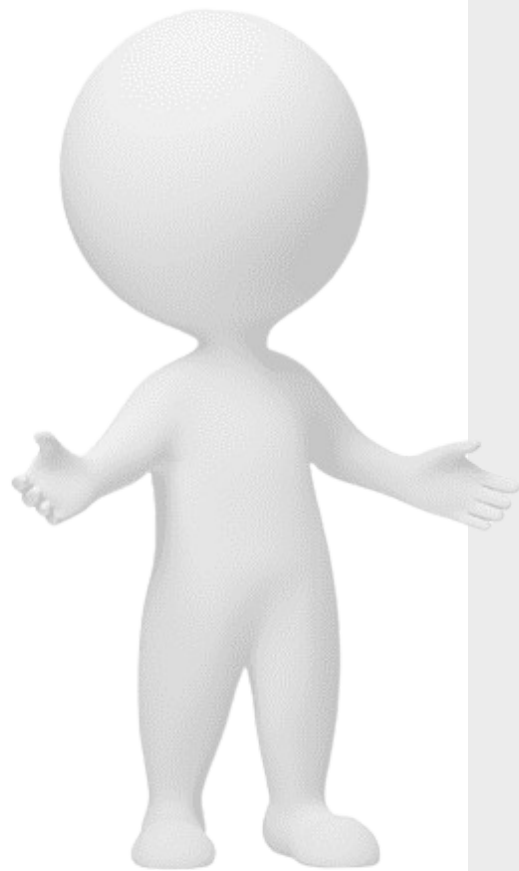


Nitya CloudTech

Dream.Achieve.Succeed

Candidate:

- **Parallelism:** Increase the degree of parallelism in ADF Copy Activities to ingest data faster.
- **Retries and Monitoring:** Enable retry policies in ADF and integrate with Azure Monitor and Log Analytics for real-time failure tracking and resolution.
- **Partitioning:** For SQL and large datasets, use source-side partitioning to split data into smaller chunks for parallel processing.
- **Integration Runtimes:** Ensure the self-hosted runtime is scaled to match ingestion workloads.
- **Throughput Optimization:** Optimize Data Lake and Synapse settings, such as file sizes and caching, to reduce downstream processing latency.



Interviewer:

How would you secure the pipeline and ensure compliance with standards like GDPR?



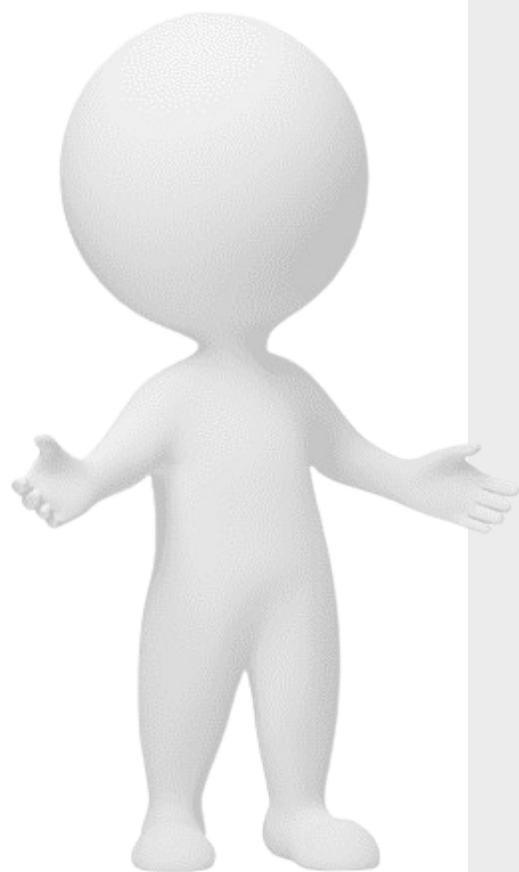


Nitya CloudTech

Dream.Achieve.Succeed

Candidate:

- **Data Encryption:** Enable encryption at rest in Data Lake and Synapse using Azure-managed keys or customer-managed keys (CMK).
- **Access Control:** Use Azure RBAC to ensure only authorized users can access data and pipeline configurations.
- **Data Masking:** Apply dynamic data masking or pseudonymization to sensitive fields, such as personally identifiable information (PII).
- **Private Endpoints:** Use Azure Private Link to ensure data does not traverse the public internet.
- **Auditing and Monitoring:** Implement activity logs and Azure Policy to enforce compliance standards across services.



Interviewer:

Suppose the analytics team complains about slow query performance in Synapse. How would you investigate and resolve this?

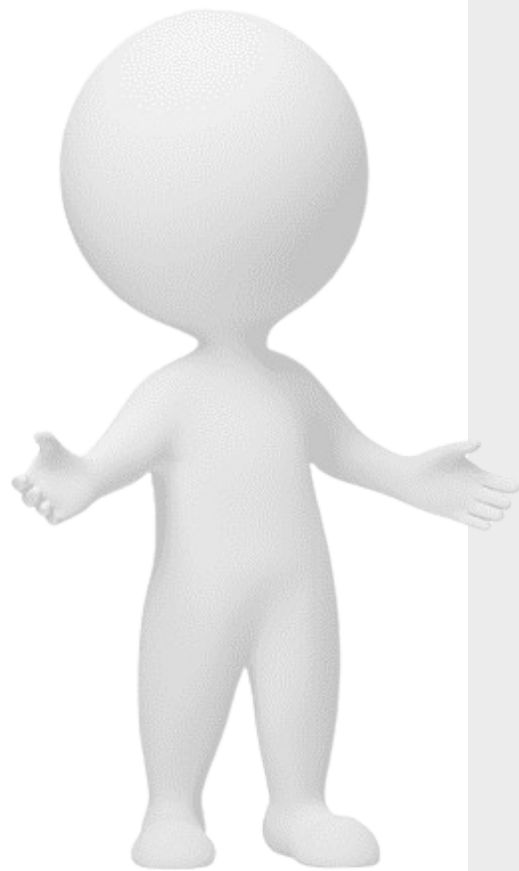




Nitya CloudTech

Dream.Achieve.Succeed

- **Query Analysis:** Use the Query Performance Insight in Synapse to identify long-running queries and their execution plans.
- **Indexing:** Ensure proper indexing and statistics updates on frequently queried columns.
- **Distribution Strategy:** Evaluate the table distribution (hash, round-robin, or replicated) and adjust for better parallelism.
- **Materialized Views:** Create materialized views for pre-aggregated datasets.
- **Caching:** Use Result Set Caching to reduce query response times for repeated queries.



Interviewer:

If the pipeline needs to process real-time data in addition to batch data, how would you extend the design?



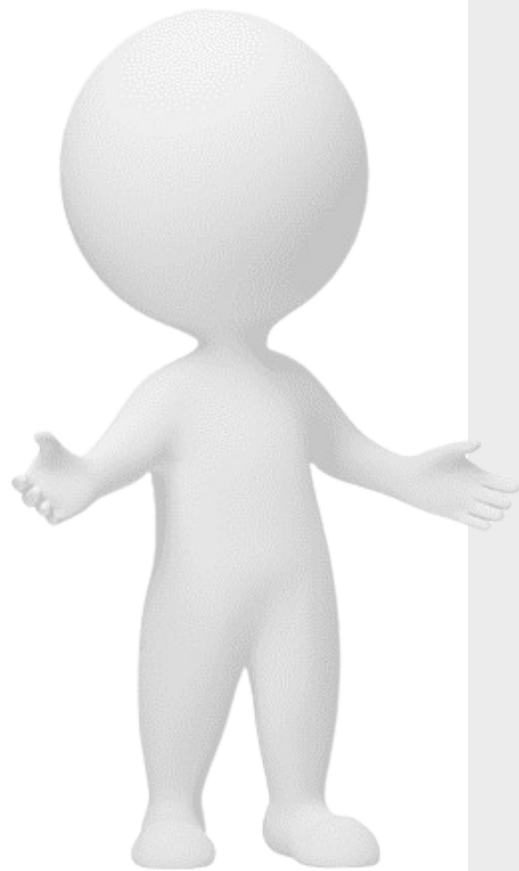


Nitya CloudTech

Dream.Achieve.Succeed

I would incorporate Azure Stream Analytics or Azure Databricks Structured Streaming:

- Use Azure Event Hubs or IoT Hub to ingest real-time data.
- Process the data using Stream Analytics queries or Databricks Structured Streaming, applying filters, aggregations, and joins as needed.
- Write the processed real-time data into Delta Lake for a unified view with batch data.
- Integrate Power BI for real-time dashboarding using DirectQuery or streaming datasets.
- This hybrid design ensures we can handle both real-time and batch processing seamlessly.





Nitya CloudTech
Dream.Achieve.Succeed

***FOR CAREER GUIDANCE,
CHECK OUT OUR PAGE***



www.nityacloudtech.com



Nitya CloudTech

Dream.Achieve.Succeed